

Unity requires physically enforced agreement

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Abstract

We prove that a content read through a noise margin, under continuous physical dynamics, depends on the graded state of no region, and we derive how far a physical coupling enforces agreement across a noisy sheet of coupled oscillators: coupling whose strength falls off more slowly than r^{-4} keeps agreement at every distance, and coupling of finite second moment, short-range in effect, lets it fade without limit. These results separate two ways in which many local descriptions of one situation can agree. In one, a coupling ties the graded state of each region to the regions it overlaps, so that any change in one, however small, pulls the others toward agreement, and the regions are parts of one physical state. In the other, found in a clocked digital chip, the parts are separate records, its bits, each held in place by a noise margin that makes it ignore every change smaller than the margin, and agreement is computed by a rule applied at each clock step. By the theorem, content read from bits is never physically enforced, whatever algorithm the bits run. Thermal noise leaves this intact. A small change in one region reaches a digital content only through the chance of a bit error, far below the content's own fluctuation, while a single weak synaptic input moves a cortical spike by

more than its jitter. Human conscious experience is unified, and we argue that unity requires its agreement to be physically enforced. The theorem is short because the definitions carry its weight, and we argue for where they place the criterion. The premise predicts that unity follows graded coupling when coupling and information flow are separated experimentally, and we state the outcomes in cortex that would refute it.

1 Introduction

A conscious scene is experienced as one scene. Many parts of the brain contribute to it, each describing part of the same situation, and their descriptions agree. This paper asks what kind of agreement that has to be.

Many parts can agree in two ways. In the first, the parts are tied together physically, like weights joined by springs. Each part's state is a graded physical quantity that pulls on its neighbours continuously, so no part can move, by however little, without pulling its neighbours after it. The parts are then coordinates of a single physical state, and they cannot vary independently at any scale. We call this agreement *physically enforced*. In the second, each part keeps its own record, like separate ledgers kept consistent by a clerk. A rule compares the records at discrete steps and corrects them where they differ. Each record is held in place by a margin, and below that margin the records vary independently of each other and of everything beneath them. We call this agreement *computed*. The first is one system in the sense physics uses, a single state space with coupled coordinates. The second is many systems kept consistent by a protocol. A clocked digital chip is of the second kind: its bits are separate records, and whatever agreement its program produces is computed. Our premise is that the unity of an experience is oneness of the first kind.

Premise P (physical unity). A system’s contents are unified only if each region’s contents are read from graded carriers and, within every window of the contents’ own time scale, some carrier of each region depends on the graded physical state of every region it overlaps, above the noise floor and through a coupling that contracts their disagreement.

P is a necessary condition for unity. It is itself a functional condition, since it picks out unity by what carriers do to one another, and it fixes that role at the grain of graded physical dynamics instead of at the grain of an algorithm. It therefore rejects computational functionalism about unity, the view that only the computation matters: a perfect simulation of a coupled field reproduces the field’s content-level dynamics and still fails P, because its agreement is computed. The reason to hold P is that unity decides how many subjects there are, and parts are one system when they interact at every scale above their own noise (Section 3). Physical systems are noisy, so P is read on the probability laws of contents. Graded dependence then counts above the noise floor: every sub-threshold change larger than the noise of what a carrier depends on must move the carrier’s next content by at least as much as that content fluctuates. A noisy carrier meets this near its threshold, so P asks it of some carriers within each window rather than of every carrier at every instant. P is neutral on where unified contents live: it constrains whichever regions carry them, whether they span the cortex or are confined to a posterior zone of it (Koch et al., 2016).

The paper makes three claims:

1. Content read through a noise margin, under continuous physical dynamics, depends on the graded state of no region, so content read from bits is not physically enforced, whatever algorithm the bits run. Measured by one protocol over the states each occupies, a cortical membrane passes the noise-floor test near its threshold and visits such states within every window of a percept’s time scale, while a logic node passes at none of its states (Section 3).

2. On a noisy coupled sheet, the reach of the coupling sets how fast agreement fades with distance: coupling that falls off more slowly than r^{-4} keeps agreement at every distance, and coupling of finite second moment lets it fade without limit (Section 4).
3. P makes testable predictions for cortex: a weak, sub-threshold perturbation of one region shifts the graded content decoded in an overlapping region toward agreement, and when two regions are bridged at matched information flow, unity follows the bridge that couples them gradedly (Section 7).

Every formal result in the paper is machine-checked in Lean 4, and Appendix C maps each one to its declaration.

2 Unity as agreement over a cover of contents

Think of an atlas. Each page maps one patch of ground, neighbouring pages overlap, and where they overlap they show the same roads. Pages that agree on every overlap fit together into one map of the whole territory. We treat the brain’s regions the same way: each describes part of a shared situation, and unity is the property that these local descriptions agree wherever they overlap. When they do, they determine a single global state.

Synchrony is not unity. Two regions can oscillate in phase while one encodes a stimulus at one location and the other at another, so equal phases coexist with incompatible descriptions. Unity is a property of what overlapping regions describe, and synchrony is at most a route to it.

A region’s *content* is a variable that the system itself reads, not merely one an outside observer can decode. Two conditions define it: the regions this region drives respond to it, and it determines the quantity the region represents, such as a location or an orientation. A variable that meets both is a *carrier* of content. A variable that only an observer can decode is a trace.

The supply current of a digital circuit carries enough of its logical state that a secret key can be recovered from it (Kocher et al., 1999), yet no gate of the circuit reads the current, so the current is a trace. In cortex the membrane potential and the timing of spikes are carriers, because the postsynaptic cells integrate them.

Agreement is measured between regions whose contents represent a shared variable. The regions, and the content variable of each, are fixed by this causal role and by decodability before agreement is measured, so that nobody can make the measurement trivial by choosing the regions after the fact.

This definition of unity, and the measurement of agreement the tests below use, come from a companion paper (Bobadilla-Suarez, 2026). It proves that compatible local descriptions determine one global state, shows that phase coherence limits their disagreement only through shared encoders of bounded sensitivity, and measures agreement with decoders fitted independently on two regions, counting only the agreement in excess of what their separate accuracies imply. P adds a condition on how that agreement is produced.

3 Physical enforcement versus routed agreement

The argument needs five definitions. Each is stated in words first.

A system has a physical state, its *micro state* $x = (x_i)_i$, made up of the states x_i of its regions. One step of its physics carries x to $f(x)$. A *content map* π reads each region’s content off the micro state. The content level is *closed* when the next content can be computed from the current content alone, without looking at the micro state: $\pi \circ f = g \circ \pi$ for some content dynamics g . A digital computer is built so that its logical level is closed in this sense.

A readout has a *noise margin* on a set of states when every state in the

set can be nudged a little without changing what is read, that is, when each has a neighbourhood on which the readout is constant. A bit is the standard example: any voltage well above the threshold reads as 1.

Content *depends gradedly* on a region when every change to that region, however small, can move the next content. Formally, no neighbourhood of the region's micro state leaves the next content unchanged while the other regions are held fixed.

The *overlap discrepancy* measures how much the contents of overlapping regions disagree. It *contracts* near a state when one step of f shrinks it by a factor below one throughout a neighbourhood of that state. Asking for a whole neighbourhood matters: perturbations too small to cross any threshold are then pulled back toward agreement along with larger ones.

Agreement is *physically enforced* at a state when content depends gradedly on each region there and the overlap discrepancy contracts near it. P requires this at the states the system occupies while its contents are unified.

The margin theorem. In words: if content is read through a margin, small enough physical changes never reach it. Precisely, if the micro dynamics is continuous and carries a state into a margin's operating set, the next content depends gradedly on no region's micro state. The proof is short: continuity carries a small enough neighbourhood of each region's micro state into the neighbourhood on which the readout is constant. A thresholded continuous quantity has a margin away from its threshold, and so does a finite word of such readouts, so the theorem covers any content read from bits. The definitions carry the weight of the result, and the rest of this section argues that they sit at the right level.

Why the criterion sits below the content level. At the content level a computer can imitate a coupled field exactly. A digital system running a consensus algorithm, or integrating the equations of a coupled field, has a content map whose next step shrinks the disagreement between overlapping

regions just as the field does. A criterion stated on the content dynamics would put that system in the same class as the field. P is stated instead on how content depends on graded physical state, and there the two differ. The field's content moves with every graded change in the regions it overlaps. The simulation's content is read from bits held inside their margins, and moves with none of them.

This is the simulation objection. A perfect simulation of a coupled cortical field reproduces the field's content-level trajectory and fails P, because its agreement is computed from contents and not enforced by the graded state that carries them. P denies the simulation unity, and the denial needs a reason.

Why unity tracks enforcement. Unity settles how many subjects there are. A unified experience is had by one subject and a divided one by two or more, which is what the split-brain debate of Section 7 asks of a single skull. Whatever makes contents unified therefore also decides whether they belong to one system or to several. Agreement at the content level cannot settle that count. Put the records of one program in different buildings, stretch the clerk's step to a year, or let one processor visit the records in turn: the content trajectory is unchanged while the physical system that realises it goes from one chip to several machines to one record at a time. Block's nation, whose citizens run a brain's functional organisation by radio for an hour (Block, 1978), is the same program spread over a country. A criterion stated on content-level agreement gives all of these the same number of subjects, without any physical fact to consult in doing so.

Counting systems at the noise floor. Physics counts systems by interaction: parts form one system when their state does not factor into independent coordinates. Stated that crudely, the criterion makes every chip one system, since wires, a shared supply and crosstalk couple all of its voltages, and it makes Block's nation one system through its radios. The criterion

needs a scale, and the parts supply it. A difference smaller than a part's own noise is erased by the part's own fluctuations, so the part does not hold it as a state. A difference larger than that noise is one the part holds, and the question is whether the other parts register it. Two carriers are one system when every change of one above its noise moves the other by at least the other's own fluctuation, and two systems when some such change leaves the other unmoved. This is the noise-floor reading of graded dependence given below, and the identity is proved: for carriers whose contents overlap, each depends gradedly above the noise floor on every carrier it overlaps exactly when every overlapping pair is one system in this sense. A region holds many carriers, and a noisy carrier registers another only while it sits near its own threshold, as measured below. Two regions are therefore one system when some carrier of each registers some carrier of the other, and P asks this within every window of the contents' time scale. Asking it of some carriers is strictly weaker than asking it of all: a state exists at which the first holds and the second fails. Both ends of the scale are the parts' own noise, so nobody chooses it to separate brains from chips. A chip fails it by design. A bit's margin spans 506 of its noise amplitudes, and no other bit registers a change anywhere in that range except through the error rate. By the noisy margin theorem below, once that rate is under the fluctuation and each carrier has a change above its noise inside the margin, no carrier of any region registers any other, so the weaker form fails exactly as the stronger one does, and each part is a system of its own. Its wires pass on only the changes that cross a threshold, which is the rule applied at each clock step, so between thresholds, where a working chip spends nearly all of its time, it is as many systems as it has bits. The count looks strange because ordinary usage counts by program, and one program uses all of the bits. Counted by physics, each bit is restored to its own rail by its own gates, and what keeps the bits in step is the rule the program applies, a protocol among separate records. The citizens of Block's nation likewise register each other only through the messages they exchange.

Cortex passes within every window, as shown below. P asks exactly this of the regions whose contents are unified. The premise is thus the conjunction of two more familiar ones: a unified experience belongs to one system, and regions are one system when their carriers interact at every scale their own noise lets them hold apart. It stands on the same footing as a workspace theory's identification of consciousness with broadcast, as a premise whose consequences can be tested, which Section 7 does.

Categories. Categorical contents have margins in every substrate, brains included. The dominant percept in binocular rivalry, a perceived category, or any class picked as the largest of several values stays the same on each side of its boundary. Away from the boundary, the margin theorem therefore denies it graded dependence in cortex exactly as in silicon. So P is stated on the graded variables from which categories are read, such as a location, an orientation, a time or the relative strength of competing percepts. A category meets P through the variable that carries it, and depends gradedly on a region only where that variable does. This opens no route for a digital machine. Every variable it computes, whether a number, a vector of numbers or a category, is a function of its bits, and after any number of steps that keep its bits away from their thresholds such a function depends gradedly on no region. The voltages beneath the bits are not carriers in the sense of Section 2. An observer may decode the machine's contents from them, but the machine itself reads them only through the bits, at that step and at every later one.

Noise. Real physics is noisy, and noise changes the question. A gate's chance of flipping is a smooth function of its input voltage and is never exactly zero, so the *probability* of a digital machine's next content depends, however slightly, on every region. Read on exact contents, the margin theorem would then pick out a difference lying below the noise floor of both substrates. P is therefore read on probabilities. The *response* of a content to a region at scale

η is the largest change that moving the region by at most η makes to the probability of any outcome of the next content, a total-variation distance. As the yardstick for a large response we take the one that a shift of one standard deviation produces in a Gaussian carrier, $2\Phi(1/2) - 1 = 0.38$: a response above it moves the content by as much as the content naturally fluctuates. Graded dependence above the noise floor asks the response to reach this fluctuation for every change of the region that is larger than its own noise and smaller than the change that crosses one of its thresholds. When the region moves a Gaussian carrier linearly, a larger change moves the content at least as much as a smaller one, so the criterion is met exactly when a change of one noise amplitude reaches the fluctuation. The constants are a Gaussian identity and the system's own scales, and none of them is fitted to either substrate.

The margin theorem has a noisy form. Let each realization of the noise fix an L -Lipschitz step of the micro dynamics, one that stretches no distance by more than a factor L . Let the content be constant within radius r of the noisy successor, except on a set of realizations of probability at most ε . Then a change of one region smaller than r/L moves the probability of every outcome of the next content by at most ε . For anything computed from bits, ε is the probability that some bit of the noisy successor lands within δ of its threshold, and the bound is attained.

A logic gate in numbers. On a logic node of 1 fF at room temperature the thermal noise is 2 mV (Johnson, 1928; Nyquist, 1928), and the textbook inverter's margin is 506 times that. After a change of one noise amplitude, the probability of a flip, which bounds the response, is below 10^{-55400} (Kish, 2002). The response reaches the fluctuation only within 0.059% of the margin's width from the threshold, where the bit is in the middle of switching.

A neuron in numbers. A spike time passes the same test. Suppose an input ΔV , too small to fire the cell on its own, arrives while the membrane

potential is rising through threshold at rate $\dot{V}$. It moves the spike by $\Delta V/\dot{V}$. Voltage noise σ_V jitters the spike by $\sigma_V/\dot{V}$. The shift therefore exceeds the jitter exactly when ΔV exceeds σ_V , whatever the slope, and spike generation adds little noise of its own (Mainen & Sejnowski, 1995). A unitary excitatory potential between cortical pyramidal cells averages 1.3 mV, with a coefficient of variation of 0.52 (Markram et al., 1997), and the membrane noise near threshold is 0.54 mV (Jacobson et al., 2005). Counting the input’s own variability as noise, one such input shifts the spike by 1.5 times its jitter, a response of 0.55. Every input larger than 0.63 mV passes, up to the input that fires the cell.

A membrane over the states it occupies. That estimate holds at the moment the membrane rises through threshold. Measured over every state a membrane occupies, by the protocol applied to the logic node, it holds near threshold and nowhere else. We model the membrane as a leaky integrate-and-fire cell driven by white noise, with time constants of 5 to 20 ms, spanning the in vivo high-conductance state (Destexhe & Paré, 1999; Destexhe et al., 2003), and stationary rates of 0.5 to 8 Hz, spanning the sparse rates of awake cortex (Hromádka et al., 2008). Its next content is the time of its first spike within 10 or 30 ms, and every change of one to 3 noise amplitudes, in either direction, must move that content by the fluctuation. Propagating the membrane’s probability exactly on a voltage grid (Appendix B), it sits on average 2.0 to 3.5 noise amplitudes below threshold and passes at 0.014 to 2.3% of the states it occupies. A bit restored to a rail at the logic node’s margin passes at none of its states, and its largest response over them is 10^{-53878} . No binary readout passes at any margin, because the fluctuation is the largest response a threshold under Gaussian noise can give to a change of one noise amplitude; only a graded content, such as a spike time, can exceed it.

A membrane passes at a given instant rarely and within a window often.

A stationary membrane enters a passing state within 400 ms, the integration time that precedes a conscious percept, with probability 0.085 to 0.997, including the case in which the region it depends on supplies only 1/16 of its membrane's noise variance. If two regions are joined by 10 carriers that pass independently, some carrier passes in such a window with probability at least 0.59; with 100, a window without one occurs less than once in 10^3 , and within 40 ms the probability is at least 0.59. At a single instant, even 1000 such carriers link two regions with probability as low as 0.025, which is why P is stated per window. The window bound is proved for independent carriers, and the probability that no carrier links two regions falls geometrically with their number. Reading the spike time through a clock of 1 or 5 ms leaves the passing fraction unchanged. What separates the membrane from the bit is the distance from threshold at which each carrier is held, a few noise amplitudes against 506.

4 How far a physical coupling enforces agreement

P asks the coupling to contract disagreement between every pair of regions whose contents are unified, so the distance it must span is the extent of those regions: up to 15 cm if a scene's contents span the cortex, and about 5 cm if they are confined to a posterior zone. P can hold for a scene only if the coupling enforces agreement across that extent. The intuition is simple. If each patch of cortex is tied only to its near neighbours, agreement has to pass from hand to hand, and a little of it leaks away at every hand, so distant patches drift apart. If there are enough direct ties across long distances, agreement can hold everywhere. This section makes that precise for a noisy sheet of coupled oscillators and finds where the line between the two falls.

An electrical picture. Picture the sheet as an electrical network: each coupling between two sites is a wire, with a conductance equal to the coupling’s strength. The *effective resistance* between two sites is the resistance a meter would read between them. When the oscillators share one natural frequency, the noisy sheet is a gradient system, and in the harmonic approximation the variance of the phase difference between two sites is the noise strength times their effective resistance. Disagreement is therefore resistance: sites that are well connected, directly or through many parallel paths, agree.

Short-range and long-range coupling. How the coupling strength falls off with distance is called its *kernel*. Under short-range coupling the resistance grows as the logarithm of distance, with a coefficient inversely proportional to the kernel’s second moment, a measure of its width. The logarithmic growth is a lower bound for every kernel of bounded range, so agreement that does not fade with distance needs coupling of unbounded range. A kernel whose tail falls off more slowly than r^{-4} supplies it: such a tail bounds the resistance at every distance. Both results match classical theory. The logarithm is the Gaussian form of the fact that short-range coupling cannot give true long-range order in two dimensions (Mermin & Wagner, 1966). The threshold r^{-4} is where a random walk with the kernel’s jumps passes from returning to its start to escaping (Chung & Fuchs, 1951; Doyle & Snell, 1984), and where the plane rotator gains true long-range order at low noise (Kunz & Pfister, 1976). Appendix A gives the arguments and the classical correspondence.

What a finite cortex can tell apart. Simulated sheets follow the harmonic identity in the ordered phase, and the kernel’s width sets both the order–disorder crossover and the rate at which agreement fades (Appendix A). Across a cortex of finite size, a wide short-range kernel and a slowly falling tail therefore differ only by a slowly growing logarithm. Telling them apart takes a measurement of the kernel’s second moment as well as of its tail

exponent.

Three facts that need no approximation. First, adding coupling never raises the effective resistance, so a long-range tail added to a short-range kernel can only improve agreement, whatever its shape. Second, a direct coupling between two sites caps their resistance at its reciprocal, at any distance. Third, a Lipschitz encoder, one whose output changes by at most a fixed multiple L of any change in its input, carries phase agreement over to content agreement: the mean-square content discrepancy is at most L^2 times the mean-square phase difference. Combined with the harmonic identity, an encoder of Lipschitz constant L at noise strength D has root-mean-square content discrepancy at most $L\sqrt{D R_{\text{eff}}}$ between two sites of effective resistance R_{eff} .

Content can agree where phases differ. Picture a travelling wave sweeping across the sheet. Every site has a different phase, yet every site sees the same local pattern: the same wave passing at the same speed. An encoder that reads the phases of a small neighbourhood relative to the phase at its centre reports the same content at every site of such a plane wave, whatever its gradient, and at every site of a winding on a ring or torus. For a wave plus a departure from it, the content discrepancy between two sites of a Lipschitz relative encoder is at most its Lipschitz constant times the departure's relative readings at the two sites. The gradient of the wave and the distance between the sites are absent from it. Travelling waves and windings, the configurations that attain the worst case of a hop-by-hop chain bound, cost a relative encoder nothing. What it pays for is the local roughness of the field.

5 The unity window

P asks the coupling to contract disagreement, and a contraction that finishes after the content has changed enforces agreement only on a content that has already been replaced. This section states when the contraction finishes in time. Agreement between two regions has a timing problem at both ends. Too early, and the influence of one region has not yet reached the other. Too late, and the content has already changed. Between the two lies a window, and unity at a given time scale requires that window to be open.

The early edge: signals must arrive. Agreement cannot arrive before the coupling’s causal cone allows. Precisely, a report at one site that tracks a variable at another to within ε , for two values of the variable more than 2ε apart, is possible after T rounds of the coupling only if the second site lies in the first site’s causal past for T rounds.

The late edge: disagreement must settle. Disagreement decays at the relaxation rate of the coupling. If the coupling shrinks the overlap discrepancy by a factor $\kappa < 1$ at every step, on a set of states it keeps, a discrepancy that starts at most d_0 is at most ε from step $\log(d_0/\varepsilon)/\log(1/\kappa)$ on. The estimate is sharp in form: a discrepancy that shrinks by at most a factor ρ per step is still above ε at every step before $\log(d_0/\varepsilon)/\log(1/\rho)$. The window opens at the later of the causal deadline and the relaxation time, and unity at the time scale of a content requires it to open before that content changes.

A worked example for cortex. The longest cortico-cortical fibres run about 15 cm (Nunez & Srinivasan, 2014). Callosal axons conduct at median velocities of 4.9 to 8.8 m/s, and the fastest of them at up to 20 m/s (Caminiti et al., 2009). A chemical synapse adds about 0.5 ms (Katz & Miledi, 1965; Sabatini & Regehr, 1996). The earliest a synaptic signal can cross the cortex is therefore 8 ms, along the fastest axons, and typical arrival takes 18 to

31 ms. For the time scale of a content, take the unconscious integration that precedes a conscious percept, which lasts up to 400 ms (Herzog et al., 2016). Count agreement as reached when disagreement has fallen 10-fold, decaying exponentially. The window is then open when the coupling’s relaxation time is at most 160 ms with typical arrival along the slower fibres, and 166 ms along the faster. Over a localized extent of 5 cm the earliest arrival is 3 ms, typical arrival along the faster fibres is 6.2 ms, and the relaxation time may reach 171 ms. Signal travel thus uses only a small part of the time available, and whether the window opens depends on how fast the coupling settles.

6 Current artificial systems

A clocked digital chip is an assortment of separate records. Each bit is held in place by its own noise margin, and below that margin it varies independently of every other bit and of the voltages beneath it. The records agree only when the program, applied at each clock step, makes them agree. A transformer running on such hardware reads every content from bits held within their margins, so the result of Section 3 applies to it: its agreement is computed, whatever it computes. Hardware whose contents are graded physical quantities coupled without barriers, such as analog, in-memory or neuromorphic substrates, can satisfy P.

Symbols and physical contents. The line P draws is between symbolic and physical content, and the line between digital and analog hardware is its most common instance. A *symbol*, in the sense the margin theorem makes precise, is a content read through a noise margin: a physical state whose identity survives every small enough change of the state that carries it. Two symbols therefore agree only when a rule applied at discrete steps makes them agree. A *physical content* is a graded quantity that the system’s own dynamics reads, so its agreement with the contents it overlaps is produced

by the coupling between them. Symbols are physical objects too; what sets them apart is the margin through which they are read. A bit is a symbol, and so is a spike count latched at a clock edge. A spike time, a membrane potential and the phase of a coupled oscillator are physical contents. P requires physical contents, and a system whose contents are all symbols computes its agreement, whatever algorithm it runs.

Organisation, not material. The criterion concerns how a substrate is organised and says nothing of what it is made of. An analog crossbar whose column currents carry content and settle through shared conductances depends gradedly on every input voltage. An array of coupled electronic or optical oscillators shrinks phase disagreement through a physical coupling exactly as a cortical sheet does. A neuromorphic chip passes where its contents are read from membrane potentials coupled continuously, and fails where they are read from spike counts latched at a clock edge. Biological tissue passes or fails on the same terms: a neural circuit whose contents were read only from all-or-none events latched far from threshold would fall under the margin theorem as any digital circuit does.

Why spikes are not bits. Spikes are all-or-none events, and what makes them graded carriers is their timing. When a membrane potential rises through threshold, the moment it crosses moves continuously with any small input, and the receiving cells integrate spike times as they arrive rather than sampling them. Spike timing therefore carries graded dependence from one region to the next. A clocked latch compares the time of a transition with the clock edge, which puts a threshold on the transition time, so the margin theorem applies with that time as the thresholded quantity. The clock alone does not separate the substrates: a spike time read through a clock still passes near threshold, as measured above. What a digital circuit adds is restoration. Each gate drives its output back to a rail, so every transition the latch reads starts and ends far from any threshold, and the clock then

removes the graded timing that spikes keep.

Clockless logic. Asynchronous circuits have no clock. In a quasi-delay-insensitive design each stage signals completion to the next through a handshake, and correctness holds under every assignment of gate and wire delays apart from a few forks assumed isochronic (Martin & Nyström, 2006). Their transition times vary gradedly with every voltage, as a spike time does. The design makes the computed values independent of those times: a value is accepted once its code is complete, whatever delays produced it. The timing of such a circuit is therefore a trace in the sense of Section 2. It moves with the physics, and the circuit’s own contents are read from completed codes, each held at a rail by restoring gates, so they meet the margin theorem as a clocked circuit’s contents do.

Leakage. Real digital hardware leaks: supply noise, crosstalk and temperature move the analog quantities behind every bit. The margin theorem says where such leakage can reach content. If a content read as a word of thresholded continuous quantities depends gradedly on any region, some bit of the next state sits exactly on its threshold, and correct operation keeps every state off that set. Digital logic does pull perturbations back, but each gate does so alone: its gain pulls a disturbed voltage back toward its own rail. That restores each bit toward its own logical value, pulls no region toward agreement with another, and is what keeps the bits inside their margins. The chip’s records stay separate even in the one place where its physics contracts.

The margin also has a width. If every bit of the next state is at least δ from its threshold, each bit is read from an L_v -Lipschitz quantity and the micro dynamics is L -Lipschitz, then a change to one region smaller than $\delta/(LL_v)$ changes no bit. Leakage below that scale reaches content only through the error rate (Section 3), and leakage above it flips a bit, which is a logical error. Digital design keeps ordinary noise well below that scale: a textbook inverter’s static noise margin is 41% of its supply voltage, against an expected

supply variation of 10% (Rabaey et al., 2003), so the expected variation is 0.24 of the margin. The nondeterminism of floating-point reduction order is a choice among logical trajectories and lies outside both theorems: it changes which content is computed, not how content depends on graded state.

The margin’s width. If P turned on how wide an engineer happened to make a noise margin, that would be a reductio of P: two machines running one program would differ in unity by a design parameter. The width is not free in that way. By the noisy margin theorem a content’s response to a region is at most its error rate, so a machine whose contents reached the fluctuation at the noise scale would err with probability at least 0.38 at every step. Its bits would fail to hold their values from one step to the next, its program’s trajectory would stop being reproducible, and whatever agreement survived would come from the physics beneath the bits and not from the program. The line P draws therefore coincides with the physical line between digital and analog operation, and working systems sit far from it on either side: a logic node’s margin is 506 noise amplitudes wide, while a cortical membrane sits on average 2.0 to 3.5 noise amplitudes below its threshold and reaches a passing state within every window of a percept’s time scale.

Attractor basins. Cortex has margins of its own. An attractor network’s basin is a dynamical margin: every state in it flows to the same attractor, so which attractor the network settles in survives every small enough change. This is the categorical content of Section 3 in dynamical form. The basin gives the attractor’s label a margin and leaves graded the state that carries the label: the position along a continuous attractor, such as the bump of activity that encodes a direction, the membrane potentials within the basin, and the spike times on the way to the attractor. Downstream cells integrate those. The voltage beneath a bit has no such reader, since the chip reads it only through the threshold. A cortical circuit whose contents were read only as attractor labels, far from the basins’ boundaries, would fall under the

margin theorem as a chip does, and the weak nudge of Section 7 tests which case cortex is in.

7 Tests

What counts as evidence about experience. Every test below measures decoded agreement: agreement of the contents available to readout. If a split in readout could always be dismissed as a split in access alone, with experience still unified behind it, no observation would bear on P. So the evidence about experience itself is fixed in advance. It consists of responses that combine stimuli across the regions in question when the task requires it, and reports of a single field of experience spanning them. A dissociation counts as evidence against P if such responses and reports split while graded coupling between the regions is intact, or stay integrated after graded coupling between them is removed.

Separating coupling from information. This is the test that decides between P and computational functionalism. In every manipulation of cortex available so far, graded coupling and information flow move together: anaesthesia and callosal section remove both. A test that confirms P over computational functionalism must change one while holding the other fixed. Stimulating the regions from outside does not separate them. Coherence between cortical regions arises largely as a consequence of communication between them (Schneider et al., 2021), and transcranial alternating current applied to two regions drives both from a common source: where in-phase stimulation raised their synchrony, it raised directed information flow with it (Reinhart & Nguyen, 2019). The coupling has to be supplied instead.

The setup is a bridge. Two regions whose natural connection is interrupted are reconnected by stimulating each in proportion to the activity recorded at the other, through one of two loops. The *analog loop*’s out-

put moves with every graded change in its input. It is analog end to end, an analog filter and amplifier driving the stimulator with no digitizing stage, because a loop that digitizes the recording before driving the stimulator would run the quantized condition in both arms. The *quantized loop*'s output is read from bits, so it has a margin.

The two loops must carry the same information and fall on opposite sides of P. Section 3 fixes what the second condition means: a change of the recorded region by its own noise amplitude must move the analog loop's output by at least the fluctuation, and the quantized loop's output by less. The analog loop passes exactly when its noise, in the region's units, is at most the region's own. The quantized loop fails at every state farther from a quantization edge than the region's noise plus 0.3 times the recording noise, so its step must be many times both. Matching information by adding noise to the analog loop defeats this. At matched information the added noise is 0.24 of the quantizer's step, so an analog loop that still passes caps the step at 4.1 noise amplitudes of the region, and the quantized loop then fails at no more than 52% of its states, taking states as uniform within a step. The analog loop's bandwidth is lowered instead, with no noise added, until the transfer entropy it carries, estimated from the recorded activity, matches the quantized loop's. The step is then free: at a step of 10 noise amplitudes, with recording noise no larger than the region's own, the quantized loop fails at 74% of its states or more. Stepping the quantized loop's resolution checks that its transfer entropy rises with the resolution, so that the resolution, and not recording noise, sets the matched value. The two loops then carry the same information, and only the analog one couples the regions gradedly.

The readout is unity, with the evidence fixed at the start of this section. Unity that follows the analog loop at matched transfer entropy supports P over computational functionalism, and unity that is the same under both loops counts against P. Intracortical stimulation in animals supports the design with decoded agreement as the measure; reported unity needs a bridged

human preparation.

A weak nudge should spread. Weak stimulation of one region, below the level that changes behaviour, should shift the content decoded in an overlapping region, in proportion to the stimulation's strength and toward agreement with the stimulated region. Transcranial alternating current and weak intracortical fields supply the nudge. The decoded content must be a graded variable, such as a location, an orientation or the strength of evidence for a percept, read from the population activity that the overlapping region's targets integrate, so that it is a carrier in the sense of Section 2. A decoded category would not do: it is constant away from its boundaries in any substrate, and would stay unchanged whether P holds or not. If the decoded content of the overlapping region stays unchanged at strengths that measurably move the stimulated one, that counts against P, since P then denies unity to the very system whose unity motivates it.

Agreement should be restored. Perturb one region and measure whether its agreement with its overlap partners comes back, and how fast. In cortex the perturbation is stimulation, and the measure is the disagreement between contents decoded from the perturbed region and from its partners. P predicts restoration at the relaxation rate of Section 5, with a restoring pull proportional to the size of the perturbation, below any threshold as well as above. The same protocol run on an artificial system, by patching the activations of one layer, head or position, has an outcome fixed in advance by the margin theorem. A digital system can restore agreement by computation, above the scale at which its logical trajectory changes, and below that scale nothing moves. Observing this checks that the hardware operates inside its margins. It is also the contrast that makes the cortical outcome informative: a restoring pull that stays graded below every threshold is the signature P attributes to enforcement.

The field’s causal cone. Brain tissue carries an electric field as well as synapses. At the frequencies of neural activity that field is quasi-static (Gratiy et al., 2017): it reaches every point of the tissue it spans with no propagation delay. Synaptic influence between distant regions, by contrast, arrives only after axonal conduction and synaptic transmission, no sooner than 8 ms across the longest cortico-cortical fibres. If two regions re-agree after a local perturbation faster than the synaptic route allows, the coupling has a non-synaptic part, and the timing measures the reach of the kernel tail of Section 4 directly.

The field’s strength sets the distance at which this test can decide anything. Endogenous fields in cortex peak near 2.4 mV/mm in vivo (Fröhlich & McCormick, 2010), networks of neurons detect fields from 0.14 mV/mm (Francis et al., 2003), and the potential of a local cortical source decays as $r^{-2.1}$ across millimetres (Rebollo et al., 2021). With the peak placed at the nearest distance of that measurement, the field of a local source falls to the detection threshold within 3.7 mm, and misses it by 5 orders of magnitude at 15 cm and by 3.5 at 5 cm. The field’s cone is therefore tested at millimetre distances, where activity has been observed to synchronize across a cut through cortex by the field alone (Rebollo et al., 2021). At centimetre distances P predicts that re-agreement arrives through synaptic coupling, which is graded as well.

The timing can be measured with intracortical recording, with care. The stimulation artefact saturates clinical amplifiers during the first milliseconds, masking the earliest monosynaptic responses (Keller et al., 2014), and early responses at distant contacts carry a volume-conducted component that scales with distance as the artefact does (Prime et al., 2020). Three measures separate active re-agreement from the passive field that carries the stimulus: template subtraction of the artefact (Trebaul et al., 2016); multi-unit activity or current-source density at the distant site, which the passive spread of the local field potential does not reach (Kajikawa & Schroeder, 2011); and a

deadline set for each subject from the tractography length of the connecting fibres (Caminiti et al., 2013).

Split brain. Cutting the corpus callosum removes the main route of both information and graded coupling between the hemispheres. Patients respond to stimuli in the two visual fields as if perception were divided, yet can report the presence of stimuli in either field with either hand. This has been read as divided perception with undivided consciousness (Pinto et al., 2017). Whether the field of experience divides is contested (de Haan et al., 2020), and on a rival switch model consciousness alternates between the hemispheres (Bayne, 2010). The remaining subcortical routes are synaptic, so they couple gradedly and carry information at once. The case therefore removes coupling and information together, which is exactly the confound the bridged preparation above is built to break.

8 Related work

Computational closure. The closest formal neighbour is the account of Rosas et al. (2024). On it, a macroscopic level that is informationally, causally and computationally closed marks emergent software in a physical system. This closure is the property Section 3 places at the content level of a digital system, and P reads it the other way round: a level closed off from the graded state beneath it is one whose agreement is computed, not enforced.

Designed to suppress its physics. Kleiner & Ludwig (2024) reach a related exclusion from a different premise. If the physical dynamics of a system is relevant to its consciousness, processors designed and verified to suppress deviations from their computational dynamics cannot be conscious. The margin theorem states that suppression formally. P confines the premise

to unity and ties it to a coupling that shrinks disagreement, which is what makes it testable on both substrates.

Biology and the medium. Seth (2026) argues that consciousness may depend on biological properties that computation does not capture. P locates the relevant property in how the substrate is organised, and admits non-biological hardware that has it (Section 6). Mortal computation, whose parameters are useful only on the hardware instance that learned them (Hinton, 2022; Ororbia & Friston, 2023), shares P’s refusal to separate the computation from the medium.

The grain of a function. Functionalism leaves open the grain at which functional roles are stated. Block (1978) showed what the coarse grain admits, a nation realizing a brain’s program, and Godfrey-Smith (2016) argues from the biological basis of cognition against the multiple realizability of mental states. P is a condition of the finer kind: the role it asks of a carrier is to register the graded state of another, which a program running on separate records does not realize.

Integrated information. Integrated information theory also denies experience to digital computers running faithful simulations of a brain (Tononi & Koch, 2015; Findlay et al., 2024). It does so through the causal structure of the hardware at the grain of its logic gates. P turns instead on how content depends on graded physical state, and on whether disagreement is pulled back.

Field theories. The electromagnetic field theory of McFadden (2020) identifies consciousness with the brain’s field, supported by evidence that endogenous fields entrain cortical activity (Fröhlich & McCormick, 2010). P treats the field as one way among several to realize graded coupling.

9 Limitations

- *The premise.* P is a premise and is not derived from the dynamics. The formal results show what follows from it, not that it holds. It is necessary for unity and not sufficient for consciousness: coupled pendulums satisfy it. The tests separate P from computational functionalism. A functionalism that states its roles at the grain of graded physical state agrees with P on every test here, and the paper does not choose between the two.
- *Hardware near its thresholds.* Under noise the margin theorem bounds a digital content's response by its error rate. Hardware run so close to its thresholds that a change of one noise amplitude moves its contents by a fluctuation would escape the bound. The thermal estimate declares the node capacitance it uses.
- *The cortical estimates.* The single-input estimate combines in vitro measurements of one connection type and one noise recording, and reads the spike-time shift through a linear threshold crossing. The occupancy result models one membrane as a leaky integrate-and-fire cell under white noise, with a threshold that has no dynamics of its own. Its time constants, rates, windows, noise shares and carrier counts are declared ranges, not fits. The window bound treats carriers as independent, and the membranes of nearby cells are correlated, which is why the carrier counts it uses stay far below the anatomical ones.
- *Reach.* The identification of phase variance with effective resistance rests on the harmonic approximation at identical natural frequencies. The logarithmic lower bound on resistance is proved for kernels of bounded range. The bound at every distance under a tail slower than r^{-4} is proved for a sheet that supplies the site counts and couplings of a dyadic chain of discs. That a kernel of finite second moment

leaves the resistance unbounded is the classical recurrence criterion, for translation-invariant kernels on the lattice; that the growth is logarithmic is established numerically on finite sheets. Above the order-disorder crossover of Section 4 the harmonic identity fails and agreement is lost at every distance.

- *Attractors.* The answer for attractor networks takes the unified contents to be carried by the graded state within a basin. A cortex whose unified contents were basin labels alone would fail P.
- *Which variables are contents.* Which variables a system's own dynamics reads, and so which are its contents, is settled system by system from its circuitry. The margin theorem takes that identification as given.
- *Artificial systems.* The predictions for artificial systems are consequences of the margin theorem. Confirming them checks that hardware runs inside its margins and leaves P untested; the tests of P are the cortical ones.
- *The unity window.* Its edges are proved for a given contraction factor; identifying that factor with the coupling's spectral gap is linear response. The relaxation time of cortical coupling is unmeasured, so the window estimate states the condition for the window to open at cortical time scales and leaves open whether cortex meets it. It also reads agreement as a declared reduction factor of disagreement.
- *From phases to contents.* The step from phase agreement to content agreement assumes one oscillator population per content patch, which is a modelling choice about cortex.
- *The field.* A long-range tail improves agreement only in proportion to its conductance against the short-range part. The field of a local cortical source falls below the detection threshold within millimetres, by an

estimate that places the field’s peak at a declared distance. Across centimetres graded coupling is carried chiefly by synapses, and the field’s contribution there is a quantity to be measured.

- *Human evidence.* The dissociation of coupling from information has, so far, no human preparation in which reported unity can be measured across a bridged connection. Human data can falsify P but cannot by themselves confirm it against computational functionalism, because graded coupling and information flow co-vary in every manipulation available so far.

References

- Bayne, T. (2010). *The Unity of Consciousness*. Oxford University Press. doi:10.1093/acprof:oso/9780199215386.001.0001.
- Block, N. (1978). Troubles with functionalism. In C. W. Savage (Ed.), *Perception and Cognition: Issues in the Foundations of Psychology*, Minnesota Studies in the Philosophy of Science, 9, 261–325. University of Minnesota Press.
- Bobadilla-Suarez, S. (2026). When does cortical coherence carry shared content? Compatibility and self-reconstruction as testable conditions. Manuscript. github.com/bobaseb/coupling_gated_cortical_coherence.
- Caminiti, R., Ghaziri, H., Galuske, R., Hof, P. R., & Innocenti, G. M. (2009). Evolution amplified processing with temporally dispersed slow neuronal connectivity in primates. *Proceedings of the National Academy of Sciences*, 106(46), 19551–19556. doi:10.1073/pnas.0907655106.
- Caminiti, R., Carducci, F., Piervincenzi, C., Battaglia-Mayer, A., Confalone, G., Visco-Comandini, F., Pantano, P., & Innocenti, G. M. (2013). Diam-

- eter, length, speed, and conduction delay of callosal axons in macaque monkeys and humans: Comparing data from histology and magnetic resonance imaging diffusion tractography. *Journal of Neuroscience*, 33(36), 14501–14511. doi:10.1523/JNEUROSCI.0761-13.2013.
- Chung, K. L., & Fuchs, W. H. J. (1951). On the distribution of values of sums of random variables. *Memoirs of the American Mathematical Society*, 6, 1–12.
- de Haan, E. H. F., Corballis, P. M., Hillyard, S. A., Marzi, C. A., Seth, A., Lamme, V. A. F., Volz, L., Fabri, M., Schechter, E., Bayne, T., Corballis, M. C., & Pinto, Y. (2020). Split-brain: What we know now and why this is important for understanding consciousness. *Neuropsychology Review*, 30(2), 224–233. doi:10.1007/s11065-020-09439-3.
- Destexhe, A., & Paré, D. (1999). Impact of network activity on the integrative properties of neocortical pyramidal neurons in vivo. *Journal of Neurophysiology*, 81(4), 1531–1547. doi:10.1152/jn.1999.81.4.1531.
- Destexhe, A., Rudolph, M., & Paré, D. (2003). The high-conductance state of neocortical neurons in vivo. *Nature Reviews Neuroscience*, 4(9), 739–751. doi:10.1038/nrn1198.
- Doyle, P. G., & Snell, J. L. (1984). *Random Walks and Electric Networks*. Carus Mathematical Monographs 22. Mathematical Association of America.
- Findlay, G., Marshall, W., Albantakis, L., David, I., Mayner, W. G. P., Koch, C., & Tononi, G. (2024). Dissociating artificial intelligence from artificial consciousness. *arXiv:2412.04571*. doi:10.48550/arXiv.2412.04571.
- Francis, J. T., Gluckman, B. J., & Schiff, S. J. (2003). Sensitivity of neurons to weak electric fields. *Journal of Neuroscience*, 23(19), 7255–7261. doi:10.1523/JNEUROSCI.23-19-07255.2003.

- Fröhlich, J., & Spencer, T. (1981). The Kosterlitz–Thouless transition in two-dimensional Abelian spin systems and the Coulomb gas. *Communications in Mathematical Physics*, 81(4), 527–602. doi:10.1007/BF01208273.
- Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may guide neocortical network activity. *Neuron*, 67(1), 129–143.
- Godfrey-Smith, P. (2016). Mind, matter, and metabolism. *The Journal of Philosophy*, 113(10), 481–506. doi:10.5840/jphil20161131034.
- Gratiy, S. L., Haldes, G., Denman, D., Hawrylycz, M. J., Koch, C., Einevoll, G. T., & Anastassiou, C. A. (2017). From Maxwell’s equations to the theory of current-source density analysis. *European Journal of Neuroscience*, 45(8), 1013–1023. doi:10.1111/ejn.13534.
- Herzog, M. H., Kammer, T., & Scharnowski, F. (2016). Time slices: What is the duration of a percept? *PLoS Biology*, 14(4), e1002433. doi:10.1371/journal.pbio.1002433.
- Hinton, G. (2022). The forward-forward algorithm: Some preliminary investigations. *arXiv:2212.13345*. doi:10.48550/arXiv.2212.13345.
- Hong, H., Park, H., & Choi, M. Y. (2005). Collective synchronization in spatially extended systems of coupled oscillators with random frequencies. *Physical Review E*, 72(3), 036217. doi:10.1103/PhysRevE.72.036217.
- Hromádka, T., DeWeese, M. R., & Zador, A. M. (2008). Sparse representation of sounds in the unanesthetized auditory cortex. *PLoS Biology*, 6(1), e16. doi:10.1371/journal.pbio.0060016.
- Jacobson, G. A., Diba, K., Yaron-Jakoubovitch, A., Oz, Y., Koch, C., Segev, I., & Yarom, Y. (2005). Subthreshold voltage noise of rat neocortical pyramidal neurones. *The Journal of Physiology*, 564(1), 145–160. doi:10.1113/jphysiol.2004.080903.

- Johnson, J. B. (1928). Thermal agitation of electricity in conductors. *Physical Review*, 32(1), 97–109. doi:10.1103/PhysRev.32.97.
- Kajikawa, Y., & Schroeder, C. E. (2011). How local is the local field potential? *Neuron*, 72(5), 847–858. doi:10.1016/j.neuron.2011.09.029.
- Katz, B., & Miledi, R. (1965). The measurement of synaptic delay, and the time course of acetylcholine release at the neuromuscular junction. *Proceedings of the Royal Society B*, 161(985), 483–495. doi:10.1098/rspb.1965.0016.
- Keller, C. J., Honey, C. J., Mégevand, P., Entz, L., Ulbert, I., & Mehta, A. D. (2014). Mapping human brain networks with cortico-cortical evoked potentials. *Philosophical Transactions of the Royal Society B*, 369(1653), 20130528. doi:10.1098/rstb.2013.0528.
- Kish, L. B. (2002). End of Moore’s law: thermal (noise) death of integration in micro and nano electronics. *Physics Letters A*, 305(3–4), 144–149. doi:10.1016/S0375-9601(02)01365-8.
- Kleiner, J., & Ludwig, T. (2024). The case for neurons: A no-go theorem for consciousness on a chip. *Neuroscience of Consciousness*, 2024(1), niae037. doi:10.1093/nc/niae037.
- Koch, C., Massimini, M., Boly, M., & Tononi, G. (2016). Neural correlates of consciousness: Progress and problems. *Nature Reviews Neuroscience*, 17(5), 307–321. doi:10.1038/nrn.2016.22.
- Kocher, P., Jaffe, J., & Jun, B. (1999). Differential power analysis. In *Advances in Cryptology — CRYPTO ’99*, Lecture Notes in Computer Science, vol. 1666, pp. 388–397. Springer. doi:10.1007/3-540-48405-1_25.
- Kunz, H., & Pfister, C.-E. (1976). First order phase transition in the plane rotator ferromagnetic model in two dimensions. *Communications in Mathematical Physics*, 46, 245–251. doi:10.1007/BF01609121.

- Mainen, Z. F., & Sejnowski, T. J. (1995). Reliability of spike timing in neocortical neurons. *Science*, 268(5216), 1503–1506. doi:10.1126/science.7770778.
- Markram, H., Lübke, J., Frotscher, M., Roth, A., & Sakmann, B. (1997). Physiology and anatomy of synaptic connections between thick tufted pyramidal neurones in the developing rat neocortex. *The Journal of Physiology*, 500(2), 409–440. doi:10.1113/jphysiol.1997.sp022031.
- Martin, A. J., & Nyström, M. (2006). Asynchronous techniques for system-on-chip design. *Proceedings of the IEEE*, 94(6), 1089–1120. doi:10.1109/JPROC.2006.875789.
- McFadden, J. (2020). Integrating information in the brain’s EM field: the cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1), niaa016.
- Mermin, N. D., & Wagner, H. (1966). Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic Heisenberg models. *Physical Review Letters*, 17(22), 1133–1136. doi:10.1103/PhysRevLett.17.1133.
- Nyquist, H. (1928). Thermal agitation of electric charge in conductors. *Physical Review*, 32(1), 110–113. doi:10.1103/PhysRev.32.110.
- Nunez, P. L., & Srinivasan, R. (2014). Neocortical dynamics due to axon propagation delays in cortico-cortical fibers: EEG traveling and standing waves with implications for top-down influences on local networks and white matter disease. *Brain Research*, 1542, 138–166. doi:10.1016/j.brainres.2013.10.036.
- Ororbia, A., & Friston, K. (2023). Mortal computation: A foundation for biomimetic intelligence. *arXiv:2311.09589*. doi:10.48550/arXiv.2311.09589.

- Pinto, Y., Neville, D. A., Otten, M., Corballis, P. M., Lamme, V. A. F., de Haan, E. H. F., Foschi, N., & Fabri, M. (2017). Split brain: Divided perception but undivided consciousness. *Brain*, 140(5), 1231–1237. doi:10.1093/brain/aww358.
- Prime, D., Woolfe, M., O’Keefe, S., Rowlands, D., & Dionisio, S. (2020). Quantifying volume conducted potential using stimulation artefact in cortico-cortical evoked potentials. *Journal of Neuroscience Methods*, 337, 108639. doi:10.1016/j.jneumeth.2020.108639.
- Rabaey, J. M., Chandrakasan, A., & Nikolić, B. (2003). *Digital Integrated Circuits: A Design Perspective* (2nd ed.). Prentice Hall.
- Rebollo, B., Telenczuk, B., Navarro-Guzman, A., Destexhe, A., & Sanchez-Vives, M. V. (2021). Modulation of intercolumnar synchronization by endogenous electric fields in cerebral cortex. *Science Advances*, 7(10), eabc7772. doi:10.1126/sciadv.abc7772.
- Reinhart, R. M. G., & Nguyen, J. A. (2019). Working memory revived in older adults by synchronizing rhythmic brain circuits. *Nature Neuroscience*, 22(5), 820–827. doi:10.1038/s41593-019-0371-x.
- Rosas, F. E., Geiger, B. C., Luppi, A. I., Seth, A. K., Polani, D., Gastpar, M., & Mediano, P. A. M. (2024). Software in the natural world: A computational approach to hierarchical emergence. *arXiv:2402.09090*. doi:10.48550/arXiv.2402.09090.
- Sabatini, B. L., & Regehr, W. G. (1996). Timing of neurotransmission at fast synapses in the mammalian brain. *Nature*, 384(6605), 170–172. doi:10.1038/384170a0.
- Schneider, M., Broggin, A. C., Dann, B., Tzanou, A., Uran, C., Sheshadri, S., Scherberger, H., & Vinck, M. (2021). A mechanism for inter-areal coher-

- ence through communication based on connectivity and oscillatory power. *Neuron*, 109(24), 4050–4067. doi:10.1016/j.neuron.2021.09.037.
- Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism. *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- Tononi, G., & Koch, C. (2015). Consciousness: here, there and everywhere? *Philosophical Transactions of the Royal Society B*, 370(1668), 20140167. doi:10.1098/rstb.2014.0167.
- Trebaul, L., Rudrauf, D., Job, A.-S., Măliia, M. D., Popa, I., Barborica, A., Minotti, L., Mîndruță, I., Kahane, P., & David, O. (2016). Stimulation artifact correction method for estimation of early cortico-cortical evoked potentials. *Journal of Neuroscience Methods*, 264, 94–102. doi:10.1016/j.jneumeth.2016.03.002.

A Resistance bounds and simulations

The logarithmic growth is a lower bound for every kernel of bounded range. Group the sites into shells around a site a , so that the coupling joins only sites in the same or adjacent shells. The shells then act as conductors in series, and the effective conductance from a to any site beyond shell n is at most $(\sum_{k < n} 1/C_k)^{-1}$, where C_k is the coupling crossing from shell k to shell $k + 1$. On a planar sheet the perimeter of shell k grows linearly in k , so $C_k \leq c(k + 1)$ for a constant c set by the kernel’s strength and width, and the effective resistance is at least $\log(n + 1)/c$. Widening the kernel enlarges its shells and raises c , which slows the growth and leaves it unbounded. Distance-independent agreement on a sheet therefore needs coupling of unbounded range.

A tail slower than r^{-4} bounds the resistance at every distance. Run a chain of site sets out from a through discs of radius 2^j and back in to b ,

the two widest discs spanning the separation. The phase difference across a and b telescopes through the discs' mean phases, and the drop between the means of two consecutive discs is bounded by the coupling energy between them, shared among the pairs of their sites. A disc of radius 2^j on a planar sheet holds at least $\alpha 4^j$ sites, and a kernel $K(r) \gtrsim r^{-(2+\sigma)}$ couples two sites of consecutive discs by at least $\beta 2^{-(2+\sigma)j}$. The number of pairs grows as 16^j and their coupling falls as $2^{-(2+\sigma)j}$, so the drops shrink geometrically with ratio $q = 2^{-(2-\sigma)/2}$, which is below one for $\sigma < 2$, and the effective conductance is at least $(1 - q)^2 \alpha^2 \beta / 8$ however far apart a and b lie.

Both results match classical potential theory. Imagine a random walker on the lattice that jumps between sites with probabilities proportional to the kernel, which is the same at every site. The resistance from a site to infinity is infinite exactly when that walker is sure to return to its start, that is, when the walk is recurrent (Doyle & Snell, 1984). Shorting every site outside a disc around a can only lower the resistance from a to a site b beyond the disc, so under recurrence that resistance grows without bound as b recedes. A symmetric kernel of finite second moment, which includes every tail steeper than r^{-4} , gives a recurrent walk in two dimensions (Chung & Fuchs, 1951). The bound above is the matching statement for tails slower than r^{-4} , whose walks escape. The same threshold holds beyond the harmonic approximation: the plane rotator in two dimensions with coupling falling as $r^{-\alpha}$, $2 < \alpha < 4$, has true long-range order at low noise (Kunz & Pfister, 1976), which short-range coupling forbids. The plane-rotator sheet also has a Kosterlitz–Thouless transition (Fröhlich & Spencer, 1981), an order–disorder crossover that the simulations below locate. When natural frequencies differ from site to site, short-range coupling fares worse still: phases then desynchronize in every dimension up to four (Hong et al., 2005).

Simulated sheets at identical frequencies, of side 64 and 128, with nearest-neighbour, exponential and power-law kernels of tail exponent $\sigma = 1$ and $\sigma = 3$ at a common row sum, follow the harmonic identity in the ordered

phase. The measured phase variance is at most 1.15 times $D R_{\text{eff}}$ at $D = 0.1$ and at most 1.26 times at $D = 0.4$, the excess rising with noise as the cosine coupling softens, and a logarithm fits its growth with distance better than a line in 16 of the 16 runs at those noise levels. At $D = 1.2$ the nearest-neighbour sheet disorders: the mean chord discrepancy $\langle 1 - \cos(\theta_a - \theta_b) \rangle$ at the largest separation is 1.00. The exponential and $\sigma = 1$ sheets stay ordered at that noise (0.46 and 0.50), and the $\sigma = 3$ sheet nearly disorders (0.92), so the kernel's width sets the order-disorder crossover as well as the rate of decay. Exact resistance sums at sheet sides 128, 512 and 1024 separate width from tail. The exponential kernel, with 25 times the nearest-neighbour stiffness, gives resistance at half the side of 0.531, 0.549 and 0.558, still growing; the $\sigma = 1$ tail gives 0.580, 0.577 and 0.577, and the $\sigma = 3$ tail, steeper than r^{-4} , gives 1.198, 1.448 and 1.572, growing as a short-range kernel does.

B The occupancy computation

The membrane is an Ornstein–Uhlenbeck process in units of its own stationary standard deviation, with the threshold at zero. After a spike it is refractory and then reset to its mean, and the mean is calibrated so that the stationary rate equals the declared rate. Probability is propagated on a voltage grid through the exact Ornstein–Uhlenbeck step, and a crossing between two sampled times is counted with the Brownian-bridge probability $\exp(-2xy/s^2)$ for sub-threshold samples x and y and step deviation s . Nothing is sampled. From each grid state, the law of the first spike time in the window, with its absence as one more outcome, is propagated in one pass, and the response to a change is the total-variation distance between two such laws. A state passes when every admissible change moves the law by at least the fluctuation, where a change is admissible if it is at least one noise amplitude of the carrier and does not carry the membrane across threshold. The fraction of occupied states that pass is the stationary mass of the passing

states. The probability of entering a passing state within a window is computed on the same chain with the passing states made absorbing. A carrier that supplies a share φ of the membrane’s noise variance has noise amplitude $\sqrt{\varphi}$ in the membrane’s units, and its changes are scaled accordingly. The bit is a carrier restored each step to a rail at a declared margin, read after one step of unit Gaussian noise, with occupancy given by that noise around the rail. Halving both the grid spacing and the time step changes the passing fraction by 10%.

C Formal results

Each formal result the paper states is a theorem of the Lean 4 development in the project repository, in its four `Phase10` modules. Table 1 maps each result to its declaration and to the section that argues from it. Every theorem rests only on Lean’s foundational axioms `propext`, `Classical.choice` and `Quot.sound`, which the build verifies for every declaration of the development. Small concrete models in `Examples/` meet the hypotheses of the bounds and show where they are attained.

Table 1: The paper’s formal results and the Lean declarations that carry them.

Result	Section	Declaration
A readout with a margin, under continuous micro dynamics, has no graded dependence on any region	3	<code>not_physicallyEnforced_of_margin</code>
The same for any content read from a finite word of thresholded continuous quantities	3	<code>not_physicallyEnforced_of_bits</code>
A thresholded continuous quantity has a margin off its threshold	3	<code>hasMargin_threshold</code>

Result	Section	Declaration
A category read from a carrier depends gradedly on a region only where the carrier does	3	<code>GradedDependence.of_comp</code>
Every function of a bit word, after steps that keep the bits off their thresholds, depends gradedly on no region	3	<code>not_gradedDependence.iterate_of_bits</code>
Shells in series bound the effective conductance	A	<code>conductance_le_shell</code>
Crossing coupling at most $c(k+1)$ gives resistance at least $\log(n+1)/c$	A	<code>conductance_mul_log_le</code>
An averaging chain from a to b bounds the resistance by $2(\sum_k \rho_k)^2$	A	<code>one_le_conductance_of_chain</code>
A tail $K(r) \gtrsim r^{-(2+\sigma)}$ with $\sigma < 2$ bounds the conductance below at every distance	A	<code>le_conductance_of_powerLaw</code>
Adding coupling never raises the effective resistance	4	<code>conductance_mono</code>
A direct coupling caps the resistance at its reciprocal	4	<code>le_conductance_of_edge</code>
A Lipschitz encoder carries mean-square phase agreement to content agreement	4	<code>integral_dist_sq_le</code>
A relative encoder reports one content at every site of a wave	4	<code>relContent_eq_of_wave</code>
A relative encoder's discrepancy is bounded by the departure from a wave	4	<code>dist_relContent_le</code>
Tracking a remote variable requires it to lie in the causal past	5	<code>mem_ball_of_reconstructs</code>

Result	Section	Declaration
Contraction by κ reaches ε by step $\log(d_0/\varepsilon)/\log(1/\kappa)$	5	<code>disc_iterate_le_of_log_le</code>
A discrepancy shrinking by at most ρ per step is above ε before $\log(d_0/\varepsilon)/\log(1/\rho)$	5	<code>lt_disc_iterate_of_lt_log</code>
Graded dependence of a bit word requires some bit exactly on its threshold	6	<code>exists_threshold_of_gradedDependence</code>
A one-region change below $\delta/(LL_v)$ changes no bit	6	<code>bits_eq_of_lipschitz</code>
Under noise, a one-region change below r/L moves the probability of every event of the next content by at most the error rate	3	<code>law_le_of_marginRate</code>
The same for anything computed from noisy bits, with error rate the probability of landing within δ of a threshold	3	<code>bits_law_le_of_lipschitz</code>
The response of a content to one region grows with the scale	3	<code>response_mono</code>
Below r/L the response is at most the error rate	3	<code>response_le_of_marginRate</code>
Graded dependence above the noise floor at every overlap holds exactly when every overlapping pair is one system	3	<code>enforcedAtNoiseFloor_iff</code>
Asking graded dependence above the noise floor of every carrier implies asking it of some carriers of each region	3	<code>EnforcedAtNoiseFloor.enforcedByCarriers</code>

Result	Section	Declaration
A state at which some carriers of each region pass and not every carrier does	3	spare_ enforcedByCarriers, spare_not_ enforcedAtNoiseFloor
Regions joined by carriers at every overlap are one system	3	EnforcedByCarriers. regionSameSystem
Bits with error rate below the fluctuation: no carrier of any region reaches any other, and each region is a system of its own	3	not_ reachesAtNoiseFloor_of_ bits, regionSameSystem_ iff_eq_of_bits
With K independent carriers each passing within a window with probability at least p , two regions go unlinked in the window with probability at most $(1 - p)^K$	3	measure_not_ reachesWithin_le
Bits with error rate below the fluctuation couple no pair of regions, so each region is a system of its own	3	sameSystem_iff_eq_of_ bits
The same bits fail graded dependence above the noise floor at any overlap of two regions	3	not_ enforcedAtNoiseFloor_ of_bits
A smaller shift of a Gaussian moves no event more than a larger one	3	gaussian_shift_le
Where the response grows with the change, graded dependence above the noise floor is decided by a change of one noise amplitude	3	gradedAboveNoise_iff_ of_mono

Result	Section	Declaration
Under linear dynamics with Gaussian noise, the response grows with the change, and graded dependence and coupling are decided at one noise amplitude	3	<code>gaussLin_</code> <code>coupledAtNoiseFloor_iff</code>